

WEB TOOL • OPEN SOURCE

From causal models to argumentation-based explanations.

The CAUSAL KNOWLEDGE BASE EDITOR is a visual interface for constructing causal knowledge bases and **reasoning** over them through **structured argumentation**, making explicit the competing causal inferences.

AVAILABLE AT
causal-knowledge-base-editor.aig.fernuni-hagen.de

WORKFLOW

1 Build a causal model

Define unobservable and explainable causal variables, connect them through Boolean causal relationships.



2 Induce an AF

Based on the causal model and observations, derive causal arguments and connect them through undercut attacks.



3 Reason & explain

Compute stable extensions, evaluate conclusions, and generate dependency- and sequence-based explanations.

FORMAL BACKGROUND

1 Causal knowledge bases

DEFINITION

Let U, V partition the atoms into **background** and **explainable** atoms. A **Boolean causal model** is a triple (U, V, K) ; a **causal knowledge base** is $\Delta = (K, A)$.

$$K = \{v \leftrightarrow \varphi_v \mid v \in V\}$$

Boolean structural equations
(one equation per explainable atom)

$$A \subseteq \text{Lit}(U), \quad \forall u \in U : A \cap \{u, \neg u\} \neq \emptyset$$

Background assumptions
(at least one literal per background atom)

DEFINITION

Given $\Delta = (K, A)$, the **induced AF** is $F(\Delta) = (\text{Args}(\Delta), \Rightarrow)$.

An **argument** is a pair $a = (\Phi, \psi) \in \text{Args}(\Delta)$:

$$\Phi \subseteq A, \quad K \cup \Phi \not\models \perp, \quad K \cup \Phi \vdash \psi,$$

$$\forall \Psi \subseteq \Phi : K \cup \Psi \not\models \psi$$

For arguments $a = (\Phi, \psi)$ and $a' = (\Phi', \psi')$, an **attack** is defined by:

$$a \Rightarrow a' \iff \exists \phi' \in \Phi' : \phi' \equiv \neg \psi$$

Causal Argument
(premises are consistent and minimal for ψ)

Undercut attack
(ψ negates a premise of a')

DEFINITION

Given $F = (\text{Args}, \Rightarrow)$, a **stable extension** is a set $E \subseteq \text{Args}$:

$$\forall a, b \in E : a \not\Rightarrow b, \\ \forall a \in \text{Args} \setminus E \exists b \in E : b \Rightarrow a$$

Stable extension
(conflict-free; attacks all outside arguments)

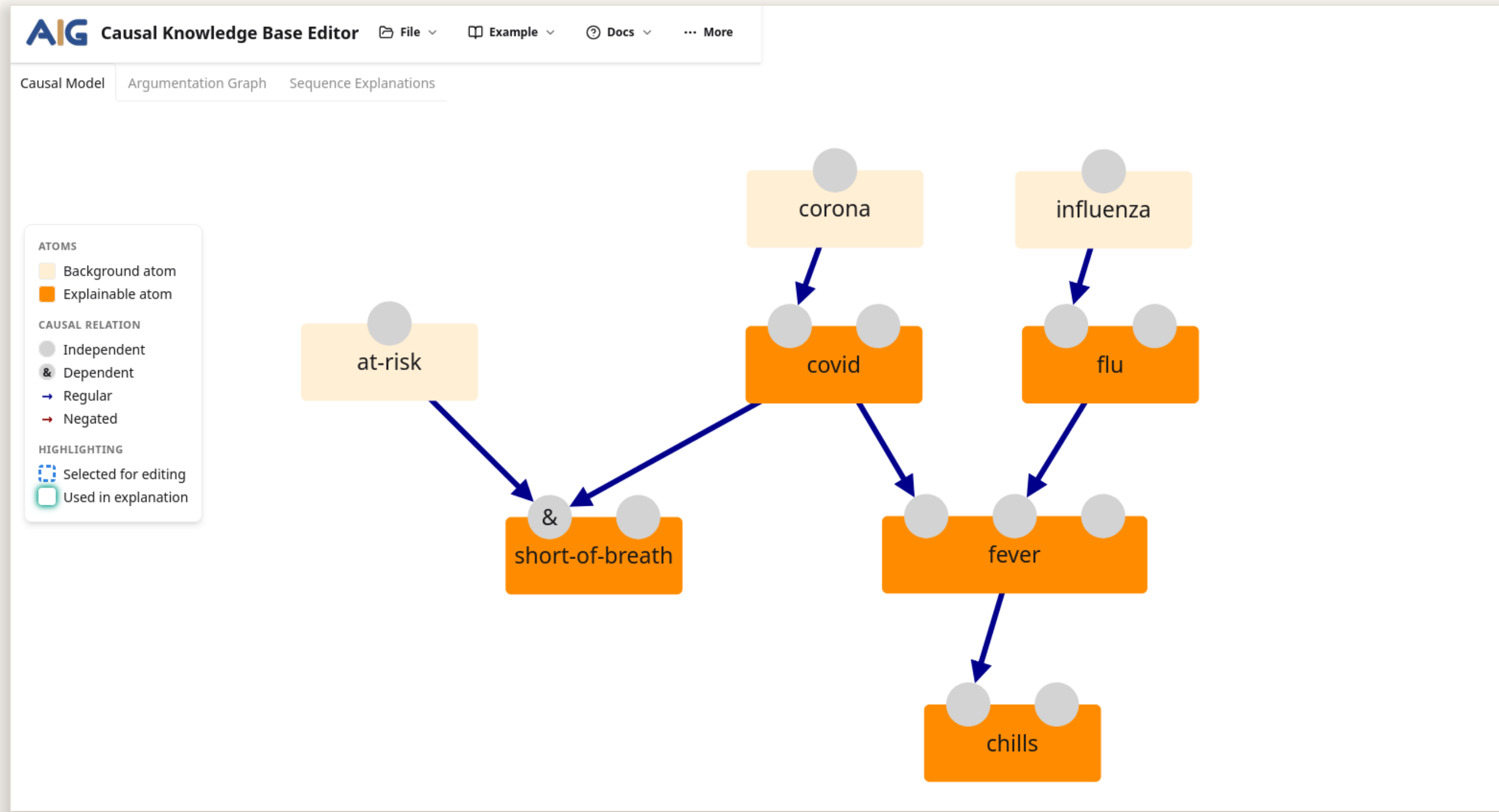
For an observation φ and conclusion ψ , **causal entailment** is defined by:

$$\varphi \vdash_{\Delta} \psi \iff \forall E \in \text{st}(F(K \cup \{\varphi\}, A)) : \\ \exists (\Phi, \psi) \in E$$

Skeptical acceptance
(ψ is supported in every stable extension)

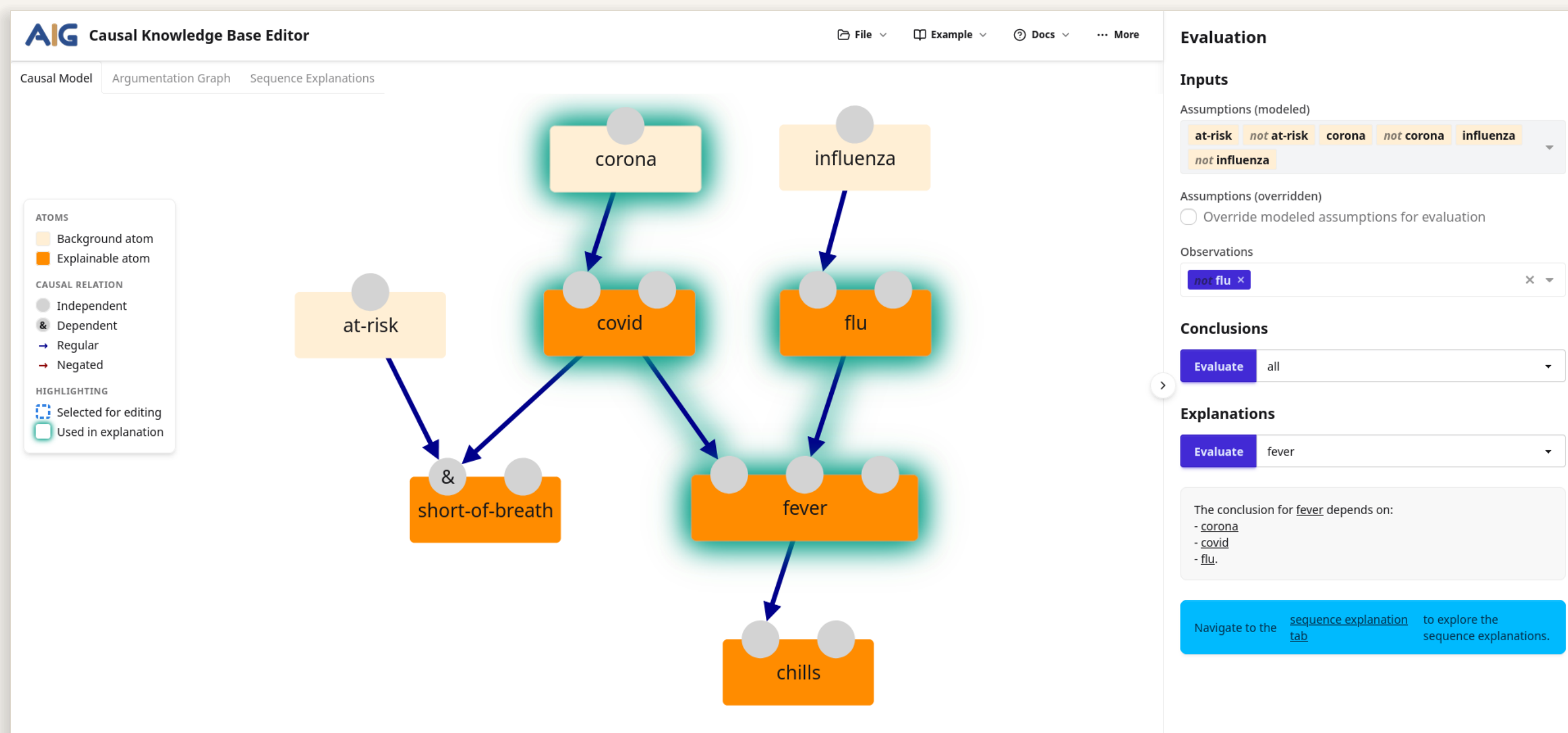
IN ACTION

1 Causal model



The example: corona causes covid, influenza causes flu, either leads to fever then chills, and covid in an at-risk patient also causes shortness of breath. **Background atoms** (light) are unobservable; **explainable atoms** (orange) are derived. A & port is a **conjunctive cause**, otherwise causes are **independent**.

3 Reasoning & explanations

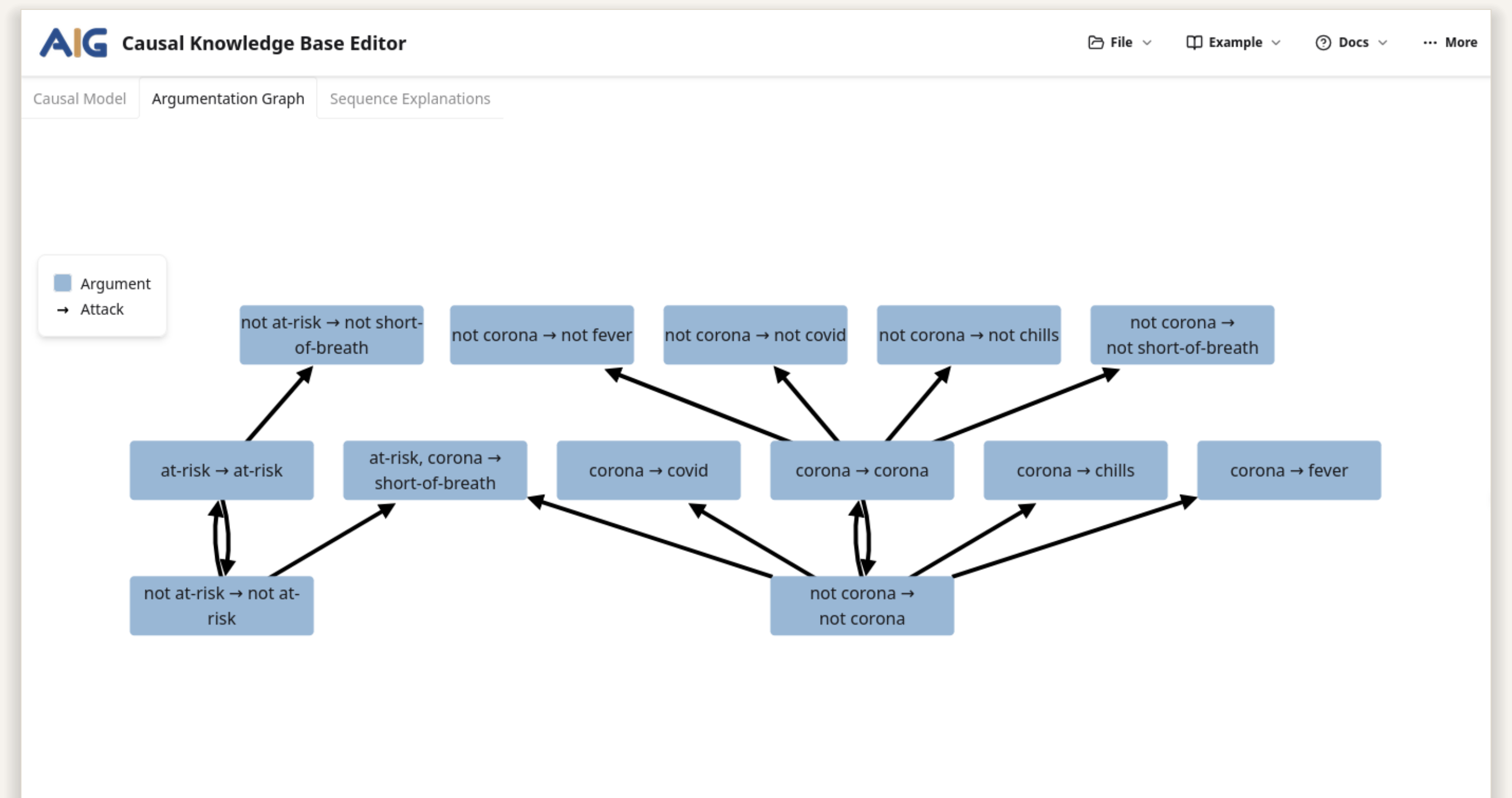


The **evaluation sidebar** lists the **causal conclusions** that follow from the model and the observation **not flu**. Highlighted nodes trace the **causal chain** deriving **fever**.

OUTLOOK

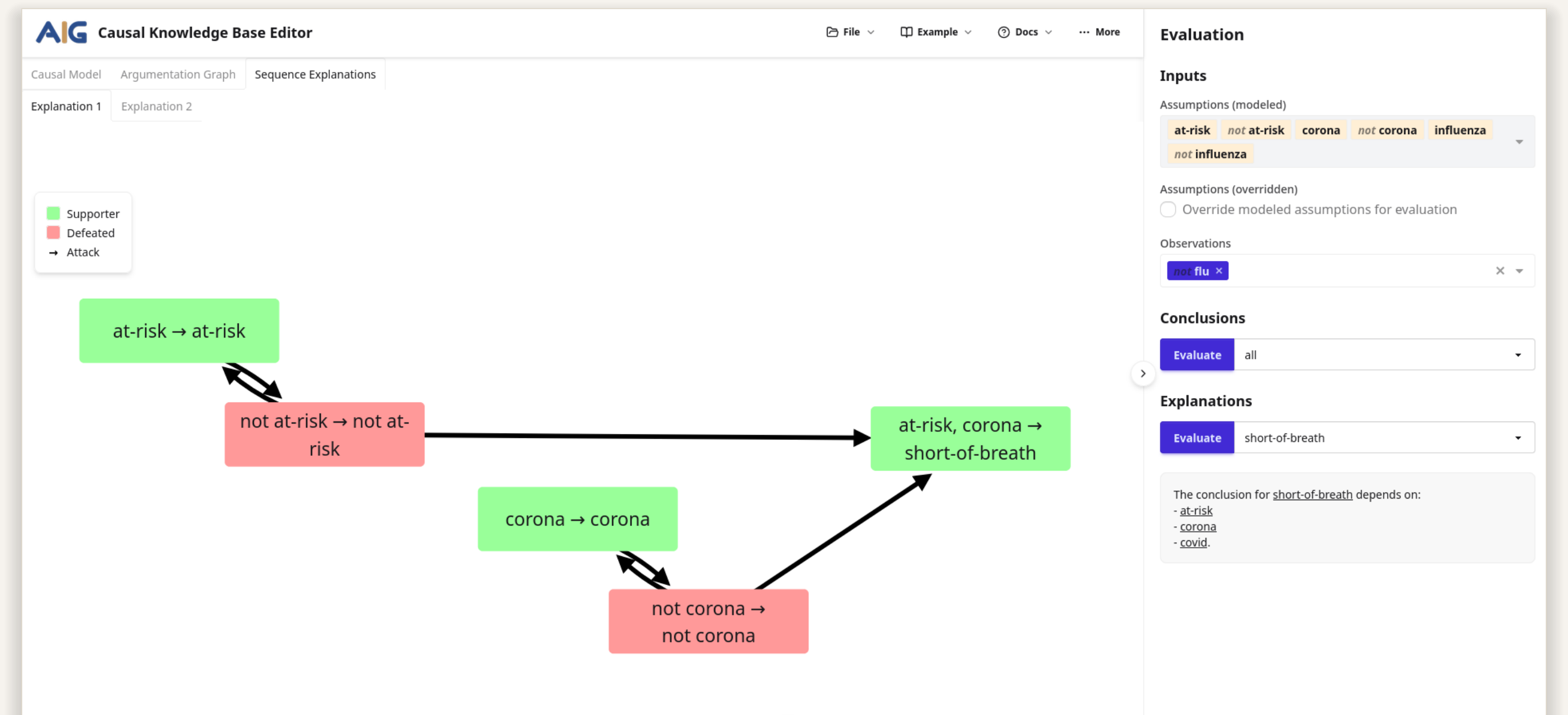
- **Presentation of Explanations.** Integrate results more tightly with the induced framework, making it easier to inspect how arguments and attacks lead to accepted conclusions.
- **Counterfactual reasoning.** Add counterfactual reasoning via the twin-model method to contrast the actual model with an intervened copy and reveal how changed conditions affect conclusions.

2 Induced argumentation framework



The AF induced by this model given the observation that the patient has **not flu**. Nodes are **causal arguments**, edges **undercut attacks**.

4 Sequence-based explanations



A **sequence explanation**: a tree-like derivation of **short-of-breath** from the **causal arguments** of the induced AF, again given the observation **not flu**.